# MLS-SBO: Mathematical Formulation

Gradient-enhanced Moving-Least-Squares surrogate-based optimization
as implemented in `scp_uno/mls_sbo_core.py`

## 1 Problem and sample archive

$$\min_x \; f(x) \quad \text{s.t.} \quad c_j(x) \le 0, \;\; j = 1, \dots, m, \qquad \ell \le x \le u. \tag{1}$$

Internally all quantities live in the normalized box $\xi \in [0,1]^d$ with $\xi = (x-\ell)/(u-\ell)$; gradients are scaled accordingly (`MLSSBOptimizer._eval`). Every true evaluation performs one FEM solve plus adjoints and stores the full tuple in the archive

$$\mathcal{S} \;=\; \big\{(\xi_i,\; f_i,\; g_i,\; c_i,\; J_i)\big\}_{i=1}^{N}, \qquad g_i = \nabla f(\xi_i) \in \mathbb{R}^d, \;\; J_i = \nabla c(\xi_i) \in \mathbb{R}^{m\times d}. \tag{2}$$

## 2 Weights and length scale

All fits use Gaussian weights centred at an evaluation point $z$ (`MovingLeastSquares._weights`):

$$w_i(z) \;=\; \exp\!\left(-\,\frac{\|z-\xi_i\|^2 - r_{\min}^2(z)}{2h^2}\right), \qquad r_{\min}^2(z) \;=\; \min_l \|z-\xi_l\|^2. \tag{3}$$

Subtracting $r_{\min}^2$ (so the largest weight is always 1) is exact — a weighted least-squares solution is invariant to a common scaling of the weights — and prevents underflow at small $h$; this is what makes the interpolation limit $h \to 0$ and linear exactness hold at *any* length scale.

**Length-scale evolution.** At iteration $k$, with trust-region radius $\delta_k$ and centre $\xi_k$ (`_min_dist_in_tr`),

$$h_k \;=\; \mathrm{clip}\big(\gamma_{\mathrm{ls}}\, d_{\min},\; h_{\min},\; h_{\max}\big), \qquad d_{\min} \;=\; \min_{\substack{i:\ \|\xi_i-\xi_k\|_\infty \le \delta_k \\ \xi_i \ne \xi_k}} \|\xi_i - \xi_k\|, \tag{4}$$

i.e. $h$ tracks the minimal distance from the trust-region centre to a sampled point inside the trust region ($\gamma_{\mathrm{ls}} =$ `ls_factor`, default 2). As the trust region shrinks and samples cluster, the fit localizes at the pace of the sampling. Only the window of the `max_points` $= 60$ samples nearest the centre enters any fit (`_fit_surrogate`), the incumbent best always retained.

## 3 Per-query gradient-enhanced linear MLS (the surrogate)

Used by the batch acquisition (§7) and as the source of the planar model. Outputs are standardized per output $o$: $\tilde{y} = (y-\mu_o)/\sigma_o$. At a query point $z$ the local linear model $q(s) = \alpha + \beta^\top s$, $s = \xi - z$, is fitted by weighted least squares to **both observation types** (Hermite MLS):

$$\begin{aligned} &\text{value rows:} && \alpha + \beta^\top s_i \approx \tilde{f}_i, && \text{weight } w_i(z),\\ &\text{gradient rows:} && \beta_l \approx \tilde{g}_{i,l}, \;\; l = 1,\dots,d, && \text{weight } w_i(z). \end{aligned}$$

Stationarity gives one shared $(d{+}1)\times(d{+}1)$ normal system for all $m{+}1$ outputs (`_solve_local`; $\varepsilon$ = Tikhonov jitter):

$$\begin{pmatrix} \sum_i w_i & \sum_i w_i\, s_i^\top \\ \sum_i w_i\, s_i & \sum_i w_i\,(s_i s_i^\top + I) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \sum_i w_i\, \tilde{f}_i \\ \sum_i w_i\,(s_i \tilde{f}_i + \tilde{g}_i) \end{pmatrix}, \tag{5}$$

where the $+I$ block is the contribution of the gradient equations. The prediction mean is $\alpha$; the *diffuse derivative* is $\beta$ (the true derivative adds weight-motion terms, see §5). Properties (test-enforced): exact reproduction of affine functions at any $h$; interpolation of the nearest sample's value and gradient as $h \to 0$.

**Uncertainty proxy.** MLS has no native variance; the acquisition uses the sample-density complement with *unshifted* weights $\bar{w}_i(z) = e^{-\|z-\xi_i\|^2/2h^2}$:

$$\sigma(z) \;=\; \sqrt{\max\Big(0,\; 1 - \textstyle\sum_i \bar{w}_i(z)\Big)} \in [0,1], \qquad \nabla\sigma \;=\; -\,\frac{\nabla\big(\sum_i \bar{w}_i\big)}{2\,\sigma}. \tag{6}$$

# 4 Exploitation model: centre-frozen anchored separable quadratic

The step model freezes the weights at the trust-region centre $\xi_k$ — the "moving" of MLS happens *across* iterations, as the centre moves — so the model is an exact polynomial whose values and gradients are mutually consistent for the SQP subproblem solver (`AnchoredSeparableQuadratic`).

**Intermediate variables** (per output $o$ and coordinate $j$; `intermediate`). With `"linear"`, $y_j = \xi_j$. With `"mma"`, a reciprocal transform is placed on the *descent* side of the anchor gradient $g^{(o)}_{k,j}$, with asymptote distance $a = \max(a_0,\, 1.3\,\delta_k)$:

$$y_j \;=\; \begin{cases} \dfrac{1}{\xi_j - L_j}, & g^{(o)}_{k,j} < 0, \quad L_j = \xi_{k,j} - a \quad \text{(lower asymptote)}, \\ \dfrac{1}{U_j - \xi_j}, & g^{(o)}_{k,j} > 0, \quad U_j = \xi_{k,j} + a \quad \text{(upper asymptote)}, \\ \xi_j, & g^{(o)}_{k,j} = 0. \end{cases} \tag{7}$$

Let $t_j(\xi) = y_j(\xi) - y_j(\xi_k)$ and $c_j = g_{k,j} / \frac{dy_j}{d\xi_j}(\xi_k)$. The model of output $o$ is

$$\boxed{m_o(\xi) \;=\; f_{k,o} \;+\; \sum_{j=1}^{d} c_{j,o}\, t_j(\xi) \;+\; \frac{1}{2}\sum_{j=1}^{d} q_{j,o}\, t_j(\xi)^2} \qquad \frac{\partial m_o}{\partial \xi_j} = \big(c_{j,o} + q_{j,o}\, t_j\big)\, \frac{dy_j}{d\xi_j}(\xi). \tag{8}$$

Since $t(\xi_k) = 0$, the anchor is exact by construction: $m_o(\xi_k) = f_{k,o}$ and $\nabla m_o(\xi_k) = g^{(o)}_k$ — the fully-linear-model requirement of trust-region theory, obtained for free because adjoint gradients are available.

**Diagonal curvature fit.** With $w_i = w_i(\xi_k)$ frozen and the gradient residuals expressed in $y$-space,

$$h_{i,j} \;=\; \left.\frac{\partial f}{\partial y_j}\right|_{\xi_i} - \; c_j \;=\; \frac{g_{i,j}}{\frac{dy_j}{d\xi_j}(\xi_i)} - c_j, \tag{9}$$

two observation types enter the weighted least squares for $q$:

$$\begin{aligned} &\text{gradient rows } (d \text{ per sample, decoupled}): && t_{ij}\, q_j \approx h_{i,j}, && \text{weight } w_i, \\ &\text{value rows } (1 \text{ per sample, coupled}): && \tfrac{1}{2}\textstyle\sum_j q_j\, t_{ij}^2 \approx r_i, && \text{weight } w_i^v = \dfrac{4\,d\,w_i}{\|t_i\|^2}, \end{aligned}$$

where $r_i = y_i^{\text{obs}} - f_k - \sum_j c_j t_{ij}$ is the value residual. The value-row weight removes the units imbalance (gradient equations are $O(t)$, value equations $O(t^2)$: raw least squares lets gradients drown values at local distances) and gives one value equation the mass of the sample's $d$ gradient equations. Value rows are enabled per output by `fit_values` $\in$ {`none`, `constraints` (default: feasibility is a statement about *values*), `all`}. Gradient-only outputs use the decoupled quotient

$$q_j \;=\; \frac{\sum_i w_i\, t_{ij}\, h_{i,j}}{\sum_i w_i\, t_{ij}^2 \;+\; \varepsilon}\,; \tag{10}$$

value-carrying outputs solve the coupled $d \times d$ normal system (once per iteration, per output), with $v_i = t_i \odot t_i$:

$$\Big[\operatorname{diag}\Big(\textstyle\sum_i w_i\, t_{ij}^2\Big) + \tfrac{1}{4}\sum_i w_i^v\, v_i v_i^\top + \varepsilon I\Big]\, q \;=\; \textstyle\sum_i w_i\,(t_i \odot h_i) + \tfrac{1}{2}\sum_i w_i^v\, r_i\, v_i. \tag{11}$$

On a separable quadratic the fit is exact (test-enforced). The bandwidth floor `min_fit_neighbors` (default 1 = tightest) can force $h$ up to the $k$-th-nearest-sample distance.

**Tangent-hyperplane surrogate (default)** (`model="tangent"`, `TangentPlaneSurrogate`). The default step model is the softmax-weighted sum of the samples' tangent hyperplanes,

$$\hat f_o(x) \;=\; \sum_i \lambda_i(x)\, T_{i,o}(x), \qquad T_{i,o}(x) = f_{i,o} + g_{i,o}^\top (x - x_i), \qquad \lambda_i(x) = \frac{e^{q_i(x)}}{\sum_l e^{q_l(x)}}, \quad q_i = -\frac{\|x - x_i\|^2}{2h^2}, \tag{12}$$

with $h$ from (4). Its gradient is available in closed form and is *exactly* the derivative of the value — moving weights included — so the surrogate is handed to the SQP subproblem as-is, with no centre-freezing and no diffuse-derivative approximation:

$$\nabla \hat f_o \;=\; \sum_i \lambda_i\, g_{i,o} \;+\; \sum_i T_{i,o}(x)\, \nabla \lambda_i, \qquad \nabla \lambda_i \;=\; \frac{\lambda_i\,(\bar s - s_i)}{h^2}, \quad s_i = x - x_i, \;\; \bar s = \sum_l \lambda_l\, s_l. \tag{13}$$

Properties (test-enforced): exact reproduction of affine functions at any $h$ (all planes coincide); interpolation of each sample's value *and* gradient in the $h \to 0$ limit (softmax $\to$ nearest-plane indicator); every new sample adds its local plane, so accumulating data refines the learned shape by construction. All outputs (objective and constraints) share the weights, so constraints learn from values and gradients alike.

**Planar variant** (`model="planar"`, `PlanarMLSModel`). The per-query MLS of (5) is solved *once* at $\xi_k$ with frozen weights, giving the affine model $m(\xi) = \hat\alpha + \hat\beta^\top(\xi - \xi_k)$ in raw units. Its value and slope blend neighbour values *and* gradients; it does not hard-interpolate the incumbent (fully-linear accuracy, which trust-region theory accepts) and carries no curvature term.

## 5 Why the model is centre-frozen (consistency)

For a *moving* fit the true derivative of the prediction is

$$\frac{d\,\alpha(z)}{dz} \;=\; \underbrace{\beta(z)}_{\text{diffuse}} \;+\; \underbrace{\frac{\partial \alpha}{\partial w}\,\frac{\partial w}{\partial z}}_{\text{weight motion}}\,, \tag{14}$$

so the diffuse part alone is *not* the derivative of the value actually returned. Handing an SQP solver the pair (moving value, diffuse gradient) gives it an inconsistent problem and corrupts its line searches — a defect measured on the GGP benchmark. Freezing $w$ at $\xi_k$ per iteration makes the model a polynomial, for which value and gradient are exactly consistent; the diffuse derivative remains only inside the heuristic batch acquisition (§7).

## 6 Subproblem, trust region, restarts

**Step subproblem** (`_solve_subproblem`); this is the whole step in the sequential regime `batch_size` $= 1$:

$$\xi^+ \;=\; \arg\min_{\xi}\; m_0(\xi) \quad \text{s.t.} \quad m_j(\xi) \le 0, \qquad \xi \in \big[\max(0,\, \xi_k - \delta),\; \min(1,\, \xi_k + \delta)\big], \tag{15}$$

solved by SLSQP with the models' analytic gradients, multistarted from $\{\xi_k,\, \xi_{\text{best}}\}$ plus two random trust-region points; fallbacks in order: phase-1 feasibility restoration $\min_\xi \sum_j \max(0, m_j(\xi))^2$, then L-BFGS-B on the penalized model merit. If $\xi^+ = \xi_k$ (model KKT point), a *null step* shrinks $\delta$ without spending a true evaluation.

**Acceptance and radius.** With the $\ell_1$ merit $\varphi(\xi) = f(\xi) + \mu \sum_j \max\big(0, c_j(\xi)\big)$ and its model counterpart $m^\varphi$, the ratio test is

$$\rho_k \;=\; \frac{\varphi(\xi_k) - \varphi(\xi^+)}{m^\varphi(\xi_k) - m^\varphi(\xi^+)}. \tag{16}$$

The centre moves when the actual reduction is positive and $\rho_k \ge \eta_a$; the radius follows the classical three-zone rule: shrink $\times 0.5$ if the step failed or $\rho_k < 0.25$; hold in the middle band; expand $\times 2$ (capped) if $\rho_k \ge 0.5$ and the step reached at least $0.8\,\delta$.

**Restarts.** On trust-region collapse ($\delta < \delta_{\min}$) or stall, restart from the incumbent best with radius cycling $\delta \leftarrow \delta_0 / 2^{(r-1) \bmod 3}$ and one randomized evaluation inside the new region (so the refit differs from the run that stalled), up to `n_resets` times: the full evaluation budget is always spent, as a sequential optimizer such as MMA would.

## 7 Batch mode (optional, $q =$ `batch_size` $> 1$)

Point 1 is the subproblem solution above; points $2, \ldots, q$ minimize the penalized lower-confidence-bound acquisition shared with GE-SBO (`gesbo_core.propose_batch`, duck-typed on the surrogate of (5)):

$$A_j(\xi) \;=\; \hat{m}_0(\xi) \;-\; \kappa_j\, \sigma(\xi) \;+\; \varrho \sum_c \max\big(0,\, \hat{m}_c(\xi) + \text{shift}_c\big)^2 \;+\; w_r \sum_{b<j} e^{-\|\xi - \xi_b\|^2 / 2h_r^2}, \tag{17}$$

with $\kappa_j = \kappa_0\, \gamma_\kappa^{\,j-1}$ an exploration ladder, $\hat{m}$ the standardized per-query MLS means, $\sigma$ the density proxy, and the last term a diversity repulsion from already-selected batch members.

## 8 Reproducibility

The optimizer is deterministic (seeded RNG, deterministic linear algebra). Bit-reproducibility of a full run additionally requires a deterministic FEM backend: the `amjax` (JAX/XLA) solver carries a per-process $O(10^{-13})$ perturbation from thread-pool reduction ordering, which the chaotic trust-region trajectory amplifies into *different local minima*; `fem_solver: direct` (scipy SuperLU) is verified bit-deterministic end-to-end.

# 9 Defaults

| Symbol | Config field | Default |
|---|---|---|
| $\gamma_{\text{ls}}$ | `ls_factor` | 2.0 |
| $h_{\min}$, $h_{\max}$ | `ls_min`, `ls_max` | $10^{-3}$, 2.0 |
| window $N$ | `max_points` | 60 |
| $\varepsilon$ | `regularization` | $10^{-8}$ |
| intermediate variables | `intermediate` | `"linear"` |
| $a_0$ | `asy_init` | 0.5 |
| value rows | `fit_values` | `"constraints"` |
| bandwidth floor | `min_fit_neighbors` | 1 |
| model | `model` | `"tangent"` |
| $\delta_0$, $\delta_{\min}$, $\delta_{\max}$ | `tr_init`, `tr_min`, `tr_max` | 0.25, $10^{-5}$, 0.75 |
| shrink / expand | `tr_shrink`, `tr_expand` | 0.5, 2.0 |
| $\eta_a$, shrink zone, expand zone | `eta_accept`, `eta_shrink`, `eta_expand` | $10^{-4}$, 0.25, 0.5 |
| $\mu$ | `penalty` | 100 |
| restarts | `n_resets` | 8 |
| $q$ | `batch_size` | 4; 1 = sequential regime |
| $\kappa_0$, $\gamma_\kappa$, $w_r$ | `kappa_base`, `kappa_growth`, `repulsion_weight` | 1, 2, 1 |